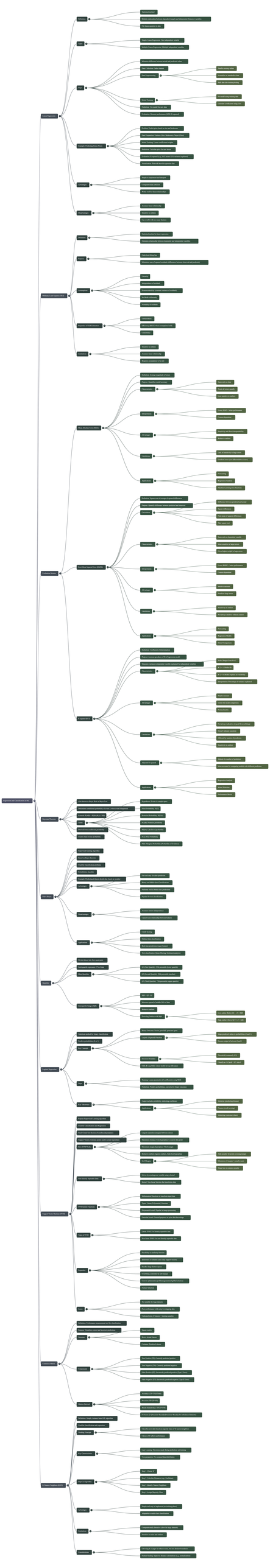

- Regression and Classification in ML
  - Linear Regression
    - Definition
      - Statistical method
      - Models relationship between dependent (target) and independent (features) variables
      - Fits linear equation to data
    - Types
      - Simple Linear Regression: One independent variable
      - Multiple Linear Regression: Multiple independent variables
    - Steps
      - Minimize difference between actual and predicted values
      - Data Collection: Gather dataset
      - Data Preprocessing
        - Handle missing values
        - Normalize or standardize data
        - Split data into training/testing
      - Model Training
        - Fit model using training data
        - Calculate coefficients using OLS
      - Prediction: Use model for new data
      - Evaluation: Measure performance (MSE, R-squared)
    - Example: Predicting House Prices
      - Problem: Predict price based on size and bedrooms
      - Data Preparation: Features (Size, Bedrooms), Target (Price)
      - Model Training: Learns coefficients/weights
      - Prediction: Calculate price for new house
      - Evaluation: R-squared (e.g., 0.85 means 85% variance explained)
      - Visualization: Plot with best-fit regression line
    - Advantages
      - Simple to implement and interpret
      - Computationally efficient
      - Works well for linear relationships
    - Disadvantages
      - Assumes linear relationship
      - Sensitive to outliers
      - Can overfit with too many features
  - Ordinary Least Squares (OLS)
    - Definition
      - Statistical method in linear regression
      - Estimates relationship between dependent and independent variables
    - Purpose
      - Finds best-fitting line
      - Minimizes sum of squared residuals (differences between observed and predicted)
    - Assumptions
      - Linearity
      - Independence of residuals
      - Homoscedasticity (constant variance of residuals)
      - No Multi-collinearity
      - Normality of residuals
    - Properties of OLS Estimators
      - Unbiasedness
      - Efficiency (BLUE when assumptions hold)
      - Consistency
    - Limitations
      - Sensitive to outliers
      - Assumes linear relationship
      - Requires assumptions to be met
  - Evaluation Metrics
    - Mean Absolute Error (MAE)
      - Definition: Average magnitude of errors
      - Purpose: Quantifies model accuracy
      - Characteristics
        - Same units as data
        - Treats all errors equally
        - Less sensitive to outliers
      - Interpretation
        - Lower MAE = better performance
        - Context-dependent
      - Advantages
        - Simplicity and direct interpretability
        - Robust to outliers
      - Limitations
        - Lack of sensitivity to large errors
        - Gradient issues (not differentiable at zero)
      - Applications
        - Forecasting
        - Regression Analysis
        - Machine Learning (loss function)
    - Root Mean Squared Error (RMSE)
      - Definition: Square root of average of squared differences
      - Purpose: Quantify difference between predicted and observed
      - Calculation
        - Difference between predicted and actual
        - Square differences
        - Find mean of squared differences
        - Take square root
      - Characteristics
        - Same units as dependent variable
        - More sensitive to larger errors
        - Gives higher weight to large errors
      - Interpretation
        - Lower RMSE = better performance
        - Context-dependent
      - Advantages
        - Intuitive measure
        - Penalizes large errors
      - Limitations
        - Sensitivity to outliers
        - Not always intuitive without context
      - Applications
        - Forecasting
        - Regression Models
        - Model Comparison
    - R-squared (R^2)
      - Definition: Coefficient of determination
      - Purpose: Assesses goodness of fit of regression model
      - Measures variance in dependent variable explained by independent variables
      - Characteristics
        - Scale: Ranges from 0 to 1
        - $R^2 = 1$: Perfect fit
        - $R^2 = 0$: Model explains no variability
        - Interpretation: Percentage of variance explained
      - Advantages
        - Simple measure
        - Useful for model comparison
        - Normed metric
      - Limitations
        - Not always indicative of good fit (overfitting)
        - Doesn't indicate causation
        - Affected by number of predictors
        - Sensitivity to outliers
      - Adjusted R-squared
        - Adjusts for number of predictors
        - More accurate for comparing models with different predictors
      - Applications
        - Regression Analysis
        - Model Selection
        - Performance Metric
  - Bayesian Theorem
    - Also known as Bayes Rule or Bayes Law
    - Determines conditional probability of event A when event B happened
    - Formula: $P(A|B) = P(B|A)P(A) / P(B)$
    - Terms
      - Hypotheses: Events in sample space
      - Priori Probability: $P(E_i)$
      - Posteriori Probability: $P(E_i|A)$
      - $P(A|B)$: Posterior probability
      - $P(B|A)$: Likelihood probability
      - $P(A)$: Prior Probability
      - $P(B)$: Marginal Probability (Probability of Evidence)
    - Derived from conditional probability
    - Used to find reverse probability
  - Naïve Bayes
    - Supervised learning algorithm
    - Based on Bayes theorem
    - Used for classification problems
    - Probabilistic classifier
    - Example: Predicting if player should play based on weather
    - Advantages
      - Fast and easy for class prediction
      - Binary and Multi-class Classifications
      - Performs well in Multi-class predictions
      - Popular for text classification
    - Disadvantages
      - Assumes feature independence
      - Cannot learn relationship between features
    - Applications
      - Credit Scoring
      - Medical data classification
      - Real-time predictions (eager learner)
      - Text classification (Spam filtering, Sentiment analysis)
  - Quartiles
    - Divide dataset into four equal parts
    - Each quartile represents 25% of data
    - Main Quartiles
      - Q1 (First Quartile): 25th percentile (lower quartile)
      - Q2 (Second Quartile): 50th percentile (median)
      - Q3 (Third Quartile): 75th percentile (upper quartile)
    - Interquartile Range (IQR)
      - IQR = Q3 - Q1
      - Measures spread of middle 50% of data
      - Robust to outliers
      - Detecting Outliers with IQR
        - Low outlier: Below Q1 - 1.5 × IQR
        - High outlier: Above Q3 + 1.5 × IQR
  - Logistic Regression
    - Statistical method for binary classification
    - Predicts probabilities (0 or 1)
    - Key Concepts
      - Binary Outcome: Yes/no, pass/fail, spam/not spam
      - Logistic (Sigmoid) Function
        - Maps predicted values to probabilities (0 and 1)
        - Ensures output is between 0 and 1
      - Decision Boundary
        - Threshold (commonly 0.5)
        - Classify as 1 if prob > 0.5, else 0
      - Odds & Log-Odds: Linear model in log-odds space
    - Steps
      - Training: Learns parameters (β coefficients) using MLE
      - Prediction: Predicts probabilities, converted to binary outcomes
    - Key Takeaways
      - Output includes probability, indicating confidence
      - Applications
        - Medicine (predicting disease)
        - Finance (credit scoring)
        - Marketing (customer churn)
  - Support Vector Machine (SVM)
    - Popular Supervised Learning algorithm
    - Used for Classification and Regression
    - Goal: Create best decision boundary (hyperplane)
    - Support Vectors: Extreme points used to create hyperplane
    - How SVM Works
      - Largest separation (margin) between classes
      - Maximizes distance from hyperplane to nearest data point
      - Maximum-margin hyperplane / Hard margin
      - Robust to outliers: Ignores outliers, finds best hyperplane
      - Soft Margins
        - Adds penalty for points crossing margin
        - Minimizes (1/margin + penalty sum)
        - Hinge loss is common penalty
    - Non-linearly Separable Data
      - Solves by creating new variable using a kernel
      - Kernel: Non-linear function that transforms data
    - SVM Kernel Functions
      - Mathematical functions to transform input data
      - Types: Linear, Polynomial, Gaussian
      - Polynomial kernel: Popular in image processing
      - Gaussian kernel: General-purpose, no prior data knowledge
    - Types of SVM
      - Linear SVM: For linearly separable data
      - Non-linear SVM: For non-linearly separable data
    - Properties
      - Flexibility in similarity function
      - Sparseness of solution (uses only support vectors)
      - Handles large feature spaces
      - Overfitting controlled by soft margin
      - Convex optimization problem (guaranteed global solution)
      - Feature Selection
    - Issues
      - Not suitable for large datasets
      - Poor performance with noisy/overlapping data
      - Underperforms if features > training samples
  - Confusion Matrix
    - Definition: Performance measurement tool for classification
    - Purpose: Visualizes correct and incorrect predictions
    - Structure
      - Square matrix
      - Rows: Actual classes
      - Columns: Predicted classes
    - Components
      - True Positive (TP): Correctly predicted positive
      - True Negative (TN): Correctly predicted negative
      - False Positive (FP): Incorrectly predicted positive (Type I Error)
      - False Negative (FN): Incorrectly predicted negative (Type II Error)
    - Metrics Derived
      - Accuracy: (TP+TN)/(Total)
      - Precision: TP/(TP+FP)
      - Recall (Sensitivity): TP/(TP+FN)
      - F1-Score: 2×(Precision×Recall)/(Precision+Recall) (for imbalanced datasets)
  - K-Nearest Neighbors (KNN)
    - Definition: Simple, instance-based ML algorithm
    - Used for classification and regression
    - Working Principle
      - Classifies new data based on majority class of 'k' nearest neighbors
      - Choice of 'k' affects performance
    - Key Characteristics
      - Lazy Learning: Decisions made during prediction, not training
      - Non-parametric: No assumed data distribution
    - Steps in Algorithm
      - Step 1: Choose 'k'
      - Step 2: Calculate Distances (e.g., Euclidean)
      - Step 3: Identify Nearest Neighbors
      - Step 4: Assign Majority Class
    - Advantages
      - Simple and easy to implement (no training phase)
      - Adaptable to multi-class classification
    - Limitations
      - Computationally Intensive (slow for large datasets)
      - Sensitive to noise and outliers
    - Considerations
      - Choosing 'k': Larger 'k' reduces noise, but less distinct boundaries
      - Feature Scaling: Improves distance calculations (e.g., normalization)